Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

REVISED SUBMISSION COPY - leader review/signature pending.

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Technical & Tools Declaration

1. Team & Project Overview

Field	Response
Team Name	奶龙 (nailong)
Selected Domain	Domain 2
SDG Track	SDG 8 - Decent Work and Economic Growth; targets 8.2 / 8.3
Project / Solution Name	Nexus
Leader	LING YUXUAN
Email & Phone (Leader)	1730211684@qq.com 1123232813
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4
Live Prototype URL	https://nexus-nailong-2026.d7ystszcjf.chatgpt.site/
Code Repository URL	https://github.com/junzhanglee/SDC2026-domain2-nailong/tree/feat/nexus-mvp

2. Architecture & Technology Stack

2.1 System Architecture Overview (maximum 200 words)

Nexus is deployed on OpenAI Sites as a Cloudflare Worker serving the frontend and backend endpoints. Browser-side rules check synthetic stock, staff availability and approved leave. Demo roles review proposals before linked tasks and activity records are created. Business records remain in browser localStorage.

For AI assistance, customer service explicitly submits one selected message to a same-origin backend endpoint. The backend requires ChatGPT sign-in, enforces daily usage limits in Cloudflare D1 and calls OpenAI Responses API using a server-side secret. Structured outputs and source evidence are validated before editable suggestions are returned. AI does not send replies, approve requests or create tasks automatically. Live verification reached OpenAI but was blocked by insufficient API credits.

Excel/CSV employee tables are read locally in the browser. WhatsApp and Microsoft 365 cloud synchronisation are not connected. Replit sponsor-provided premium access was used for development and testing; it is separate from the current Sites deployment.

2.2 Languages, Frameworks & Libraries

Name & Version	Purpose in Project	Licence
HTML5	Accessible page structure and forms.	N/A - web standard
Modern CSS	Responsive layouts and glass effects.	N/A - web standard
JavaScript ES Modules (ES2015+)	UI, local business rules and workflow.	N/A - language standard
Node.js v24.19.0	Local tests, build tools and preview server.	MIT; bundled components retain their licences
fflate 0.8.3	Browser-local ZIP decompression for .xlsx imports.	MIT; full notice in THIRD_PARTY_LICENSES/fflate-MIT.txt

3. APIs & Cloud Services

3.1 External APIs

API & Provider	Purpose in Project	Access Tier
OpenAI Responses API - GPT-4.1 mini	Selected-message requirement extraction and editable reply drafts. Adapter deployed; successful generation blocked by insufficient API credits.	Paid, pay-as-you-go; not sponsor-provided

No live WhatsApp, payment, CRM, payroll, commerce or Microsoft Graph API connection is claimed. Local Excel file reading is not an external API.

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
OpenAI Sites	Hosting, deployment and server-secret configuration.	Managed; plan/region not verified
Cloudflare Workers via Sites	Backend endpoints and server-side OpenAI requests.	Platform-managed; region not verified
Cloudflare D1 via Sites	Daily AI usage counters only.	Platform-managed; location not verified
ChatGPT sign-in via Sites	Identity before AI requests; not enterprise membership.	Platform-managed; region not verified
Replit	Project import, development and testing.	SDC-provided premium access; region not verified
GitHub / GitHub Actions	Public code hosting and configured test/build workflow.	Repository public; plan/region not verified

The prototype is deployed through Sites as an ESM Worker containing the frontend assets and backend endpoints. Replit was used for development/testing; a public Replit deployment has not been verified. GitHub Actions is configured; the 61-test result reported here was locally verified.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

Model & Version	Provider	Usage	Licence / terms
GPT-4.1 mini (gpt-4.1-mini)	OpenAI Responses API	Pre-trained; no fine-tuning or self-training.	Proprietary; OpenAI API service terms.

Status: integration deployed; live verification returned credit_balance_exhausted. No successful model generation is claimed.

4.2 Datasets

Dataset	Source	Type	Licence	Personal data?
Nexus example business records	Team-created; apps/web/assets/data.js and customer-service.js	Synthetic	Apache-2.0	N - fictional records
Workflow / file-import fixtures	Team-created; tests/	Synthetic	Apache-2.0	N
Entered messages and Excel/CSV content	User-provided during use	Private / user-provided	User-owned; permission required	Y - may include names or messages
Identity and AI usage counters	Sites identity and D1 ai_usage	Private / app-generated	Provider terms and privacy requirements	Y - platform identifiers

No project datasets are used to train or fine-tune the model. Runtime demonstrations and tests use synthetic records; users are instructed to avoid sensitive real data.

4.3 AI-Assisted Development Tools

Yes. Generative-AI tools assisted project development.

Tool & Version	Used For	Extent
OpenAI ChatGPT; hosted version not recorded	Product planning, requirements, design guidance and text drafting.	Major
OpenAI Codex; desktop version not recorded	Code generation, debugging, tests, integration, deployment and documents.	Major

Other team tools, including any Replit Agent use, require team verification; Replit account access alone is not evidence of using its AI agent.

Review, Testing and Responsibility (maximum 100 words)

Our team refines AI-generated code and text against project requirements. Verification includes 61 automated tests and browser checks of business rules, demo permissions, approvals, AI failure handling and Excel imports. Documentation distinguishes implemented functions, simulations and future work. Incomplete integrations and the API credit limitation are disclosed. Final review, submission accuracy and responsibility remain with the team; the leader must review and sign the official copies.

5. Sponsor-Provided Resources

Yes - Replit official premium access was supplied through SDC, as confirmed by the team.

Sponsor	Resource	How It Was Used
Replit	SDC-provided premium account access	Imported and ran Nexus in a cloud development environment; supported development and testing.

Essential or Replaceable? (maximum 100 words)

Replit's SDC-provided premium access supported cloud development and testing. It was helpful but replaceable: the project can also run locally and is currently deployed through OpenAI Sites. The core workflow does not depend on Replit-specific services. RunPod was not used; the applicability of its required proof-of-use item must be confirmed with the organisers.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy (maximum 150 words)

Nexus stores entered names, profiles, messages, requests and activity records in browser localStorage until users clear site data. Excel/CSV files are processed locally and are not uploaded. AI requests send only the selected message and Malaysia date to OpenAI after an explicit user action. API keys remain in server secrets. AI access requires ChatGPT sign-in; platform user identifiers and daily usage counters are stored in Cloudflare D1. These counters currently have no automatic deletion schedule. Requests use HTTPS and store:false, which does not guarantee zero provider retention. Frontend role filtering is for demonstration, not production-grade authorisation. Users are instructed to use synthetic data and avoid sensitive information.

6.2 Known Limitations & Risks (maximum 150 words)

AI may misinterpret messages, invent details, produce biased wording or follow malicious instructions. Mitigations include structured outputs, source-evidence validation, missing-field handling and mandatory human review. AI cannot automatically send replies, approve requests, make payments or create tasks. Authentication, input limits and daily quotas restrict API access. Provider failures retain the original message and show an error. OpenAI is currently the sole model provider, and insufficient credits prevent successful generation verification. Business records remain browser-local, with no cross-device synchronisation or production tenant isolation. WhatsApp and Microsoft 365 cloud synchronisation are not connected.

7. Compliance & Readiness Checklist

[ ] All resources and any additional member tools are confirmed by the team.
[x] Known third-party components are declared; fflate attribution is included in the repository.
[x] This revised document contains no API keys or passwords; the local key file is Git-ignored and hosted secrets are server-only.
[ ] The team has completed its final review of all AI-assisted outputs.
[x] Confirmed official Replit sponsor access is disclosed in Section 5.
[x] Public prototype and application-branch links are provided; the AI credit limitation is disclosed.
[ ] Leader to confirm all final fields, word limits, required evidence and signature before submission.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team knowledge, and that every API, cloud service, AI tool, dataset, model and sponsor resource used in the project has been declared.

Leader review and signature are still required. No signature is applied by this revision.

Field	Response
Leader	LING YUXUAN
Signature	Leader to sign after final review.
Prepared on	24 October 2026

Appendix A - Current End-to-End System Blocks

Path	Connected components
Business workflow	Customer/support UI -> confirmed fields -> local stock/staff/leave rules -> human approval -> tasks and local activity records.
AI assistance	Explicitly selected message -> Sites Worker -> ChatGPT identity and D1 quota check -> OpenAI Responses API -> validated, editable draft -> human review.
File import	Local .xlsx / .csv -> browser parser (fflate for .xlsx) -> field and row validation -> employee-resource summary. File bytes are not uploaded.
Cloud runtime	Sites deployment -> Cloudflare Worker serving UI and API; D1 persists usage counts only; OPENAI_API_KEY remains in server secrets.
Current limits	API balance blocked successful generation. Business roles and records remain browser-local. WhatsApp and Microsoft 365 synchronisation are unconnected.

Team Members

Name	Confirmed capacity
LING YUXUAN	Team leader
LI JUNZHANG	Team member
XIE WEIYE	Team member
HU ZHAOMING	Team member

The two official PDF templates do not ask for individual duty assignments. No unconfirmed duties are invented here. The repository may separately record team roles once confirmed.

References and Submission Status

United Nations SDG 8: https://sdgs.un.org/goals/goal8
Official repository template: https://github.com/michaelzhngg/SDC_Hackathon_Submission_Template
Application branch: https://github.com/junzhanglee/SDC2026-domain2-nailong/tree/feat/nexus-mvp
Live prototype: https://nexus-nailong-2026.d7ystszcjf.chatgpt.site/

Outstanding submission actions: leader review/signature; actual Replit proof-of-use evidence; slide-deck and demo-video links; organiser confirmation of RunPod proof applicability. This review copy does not fabricate these items.